



FOR JAVA DEVELOPERS MOVING TO NESTJS

# Changed. Not saved.

## Spring Boot to NestJS, episode 13: TypeORM entities and repositories

The annotations look the same. The lifecycle is not.

JPA tracks managed entities: change one inside a transaction and Hibernate writes it at flush. TypeORM persists when you ask.

TypeORM gives you two ways to ask, `save()` and `update()`, and they are different paths.

A Java Long is not a TypeScript number, and some TypeORM advice depends on the major version.

### In scope

- Entity and repository mapping, JPA to TypeORM
- Managed entities against plain objects
- `save()` against `update()`, measured
- bigint, and the 0.3 to 1.x change in where values

### Not in scope

- Migrations and relations: next episode
- Transactions: episode 15

Measured on one machine against Postgres 18.

## 01 The measurements

|                                |                                                   |
|--------------------------------|---------------------------------------------------|
| LOAD, CHANGE, COMMIT, NO SAVE  | Spring: written. TypeORM: not written             |
| save(entity)                   | SELECT, transaction, UPDATE; @BeforeUpdate ran 1x |
| update(id, partial)            | one UPDATE; @BeforeUpdate ran 0x                  |
| bigint 9007199254740993        | string; Number() gives ...992; BigInt exact       |
| findOneBy({ id: undefined })   | 0.3.30 returned a row; 1.1.1 throws               |
| synchronize, a property rename | RENAME COLUMN, data kept (one case)               |

Same table shape on both stacks.

## 02 The mapping transfers

@Entity stays @Entity, @Id with a generated value becomes @PrimaryGeneratedColumn, @Column stays @Column. A TypeORM Repository is an EntityManager scoped to one entity type.

probes.ts, the TypeORM entity

```
@Entity('ep12_accounts')
class Account {
  @PrimaryGeneratedColumn()
  id!: number;

  @Column()
  owner!: string;

  @Column({ type: 'bigint' })
  balanceInMinorUnits!: string;
}
```

Hibernate wrote the changed owner with no save() call; TypeORM did not

Run on a real machine, 2026-09-30, by scripts/verify-typeorm.sh in the course repository.

## 03 Ask explicitly, and pick the path

save() works from the entity: it re-read the row, opened a transaction, updated, and ran the @BeforeUpdate listener. update() sent one statement and the listener did not run.

probes.ts

```
loaded.owner = 'dave-via-save';
await repo.save(loaded);

await repo.update(dave.id, { owner: 'dave-via-update' });
```

synchronize: "Be careful with this option and don't use this in production - otherwise you can lose production data."

TypeORM 1.1.1 source and changelog, read 2026-09-30.

1.0.0 changelog, invalid-where-values-behavior: "make throw the default"

TypeORM 1.1.1 source and changelog, read 2026-09-30.

## Before you port a JPA service to TypeORM

1. Call `save()` or `update()`; a changed entity is not written by itself
2. Use `save()` when entity listeners must run
3. Map Java Long to string or BigInt, not number
4. Check the TypeORM major version before trusting old advice
5. Keep synchronize off outside development
6. Move schema changes to migrations

### Everything else

|                                                                                                                                                                                                                            |                                                                                                                                                                                                       |
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Technical claims verified 2026-09-30 against the documentation named beside each one. Vendors change behaviour and defaults between versions, so check the version you are actually running before you rely on any of it.